THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 7 & 8: **A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 50°	angle sum is 180°
2	**(d)** 54°	angle sum is 180°
3	**(c)** 40°	exterior angle = sum of the two interior opposite angles
4	**(a)** No	sum of any two sides must exceed the third
5	**(b)** 70°	base angles are equal
6	**(b)** 3/4	multiply numerators and denominators, then simplify
7	**(c)** 2/3	multiply numerators and denominators, then simplify
8	**(b)** 3/2	multiply by the reciprocal of the divisor
9	**(d)** 5/3	interchange numerator and denominator
10	**(b)** 6/13	interchange numerator and denominator

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	70°; isosceles (2).	2
12	Three (1); at the orthocentre (1).	1+1
13	5/8 m² (2).	2
14	9/20 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	60° each (1.5); exterior 120° (1.5).	1½+1½
16	7/2 = 3 1/2 acres (3).	3
17	55° (3).	3
	OR: All angles < 90° (1.5); one angle > 90° (1.5).	1½+1½
18	63 ÷ 3/2 = 42 km/h (3).	3
	OR: 10 (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Yes (1). (ii) Yes: 36 + 64 = 100 (1). (iii) 14 km (1). (iv) 4 km (2).	1+1+1+2
	OR: (i) Three (1). (ii) Centroid (1). (iii) Divides each median in the ratio 2 : 1 (3).	1+1+3
20	(i) 12 (1). (ii) 0 (1). (iii) ₹5,400 (1). (iv) ₹2,520 (2).	1+1+1+2
	OR: (i) 44/3 m² (2). (ii) ₹1,320 (2). (iii) 16 1/3 m (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapters 7 & 8: **A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 60°	180 ÷ 3
2	**(a)** altitude	forms 90° at the foot
3	**(d)** 3	one from each vertex
4	**(c)** three sides	SSS
5	**(b)** equiangular and isosceles	all sides equal
6	**(d)** 28	‘of’ means multiply
7	**(c)** not defined	cannot divide by 0
8	**(b)** 1/4	1/2 × 1/2
9	**(b)** 1	1 × 1 = 1
10	**(d)** 3/8	3/4 × 1/2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	22.7 cm (2).	2
12	52° (2).	2
13	11/4 (1); 4/11 (1).	1+1
14	4 1/2 L (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Greater than 3 cm and less than 13 cm (3).	3
16	2/3 × 11/12 = 11/18 (3).	3
17	Circumcentre (1); equidistant from the three vertices (2).	1+2
	OR: No, the sum would be less than 180° (3).	3
18	2 × 2/5 = 4/5 (3).	3
	OR: 3/10 (1.5); 1/4 (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 70° each (2). (ii) acute isosceles (1). (iii) 140° (2).	2+1+2
	OR: (i) 25.5 cm (2). (ii) 13.5 cm (2). (iii) 60° (1).	2+2+1
20	(i) Takes a part (2). (ii) More parts fit (2). (iii) 1/2 × 8 = 4; 8 ÷ 1/2 = 16 (1).	2+2+1
	OR: (i) 8/9 (2). (ii) 22/27 (2). (iii) 8/9 = 24/27 larger (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 7 & 8: **A Tale of Three Intersecting Lines, Working with Fractions** | Class 7 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 40°	angle sum is 180°
2	**(a)** 100°	angle sum is 180°
3	**(b)** 35°	exterior angle = sum of the two interior opposite angles
4	**(d)** Yes	sum of any two sides must exceed the third
5	**(a)** 65°	base angles are equal
6	**(c)** 1/4	multiply numerators and denominators, then simplify
7	**(b)** 11/8	multiply numerators and denominators, then simplify
8	**(a)** 7/6	multiply by the reciprocal of the divisor
9	**(d)** 4/7	interchange numerator and denominator
10	**(d)** 9	‘of’ means multiply

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No; sum is 190° (2).	2
12	90° (2).	2
13	Division gives 9/8, larger since divisor <1 (2).	2
14	1/12 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	360° (3).	3
16	16 cans (3).	3
17	3 m (3).	3
	OR: Draw 6 cm base (1); arcs 4 cm and 5 cm (1); join (1).	1+1+1
18	2/3 m (3).	3
	OR: Number 36; 27 (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 53° (1). (ii) QR (1). (iii) 143° (1). (iv) 8 (2).	1+1+1+2
	OR: (i) Sum would exceed 180° (2). (ii) 2, 3, 6 (1). (iii) 200° > 180° (2).	2+1+2
20	(i) 3/5 L (1). (ii) 9/10 L (1). (iii) 5 (1). (iv) 3/25 L (2).	1+1+1+2
	OR: (i) 20 (2). (ii) 4 (2). (iii) 15 = 3/4 × 20 and 4 ÷ 0.4 = 10 (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** scalene	no two sides equal
2	**(d)** median	definition
3	**(a)** 180°	angle sum property
4	**(c)** circumcentre	equidistant from vertices
5	**(d)** obtuse-angled	from the acute vertices
6	**(b)** 21	‘of’ means multiply
7	**(d)** multiplying by its reciprocal	rule
8	**(d)** 12	3 ÷ 1/4 = 12
9	**(a)** 15	5 × 3
10	**(c)** 1/4	1/8 ÷ 1/2

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No, the sum would be 180° leaving 0 for the third (2).	2
12	a + b > c, b + c > a and a + c > b (2).	2
13	2/3 (2).	2
14	7 ÷ 7/4 = 4 (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	9x = 180 (1.5); x = 20° (1.5).	1½+1½
16	9 km (3).	3
17	50° (3).	3
	OR: Two congruent right triangles formed (3).	3
18	Sold 1 1/2 kg; left 1 kg (3).	3
	OR: 9 vs 10; first smaller (3).	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Two altitudes outside (2). (ii) Outside the triangle (2). (iii) No (1).	2+2+1
	OR: (i) 3 × 180 − 180 = 360 (2). (ii) 110° (2). (iii) 130 + 120 + 110 = 360 (1).	2+2+1
20	(i) Read 80; rest 280; then 105; left 175 (3). (ii) 175/360 = 35/72 (2).	3+2
	OR: (i) 750 m (2). (ii) 24 min (2). (iii) 625 g (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.